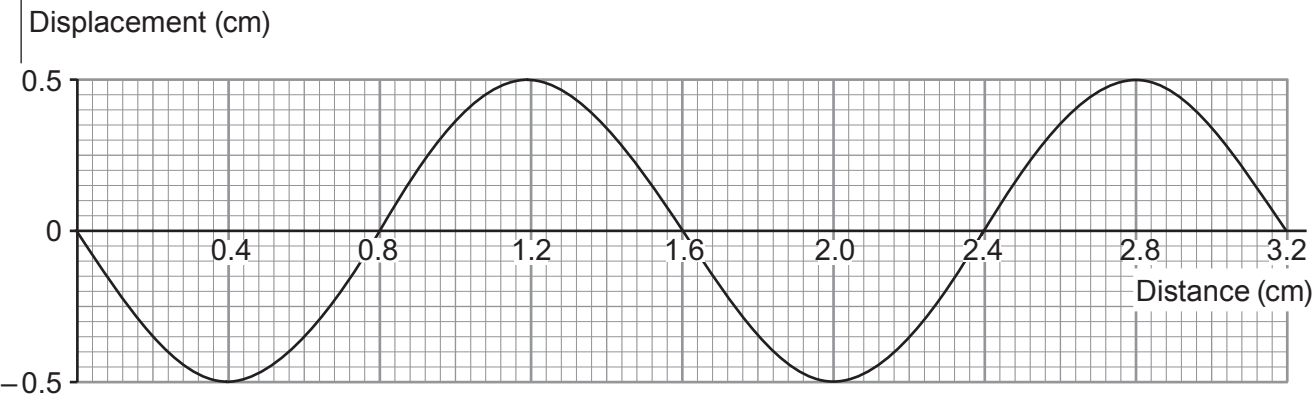


Answer all questions.

1. A teacher demonstrates water waves in a ripple tank. The diagram represents the waves.



- (a) (i) State the wavelength of the wave. cm [1]
- (ii) State the amplitude of the wave. cm [1]
- (b) It is observed that 10 waves pass a point in the tank in 2 s.
- (i) Underline the correct value for the frequency of the waves.
- 2 Hz 5 Hz 10 Hz 20 Hz [1]
- (ii) Select an equation from page 2 and use it to calculate the wave speed. [3]

Wave speed = cm/s



(c) The waves are made to reflect off a plane barrier. **Complete the diagram** to show the reflected wavefronts. [2]

